

# FINAL EXPLANATION: CORE MATHEMATICS GRADE 10

## *Student Assessment Document*

### FINAL EXPLANATION: CORE MATHEMATICS GRADE 10 – SUB-STRAND 2.3 ROTATION – FINAL EXPLANATION

#### Student Assessment Document

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| Student Name | _____ |
| Class        | _____ |
| Date         | _____ |

#### INSTRUCTIONS FOR STUDENTS

You have been studying the minute hand of the clock on the Administration Block at the University of Nairobi. Answer the driving question: When the minute hand completes its journey around the clock face, what mathematical rules govern exactly where every point on the hand ends up, and how can we use those rules to rotate any shape to any position, on paper and on a grid? Work through the five sections below. Write in full sentences, show every step of your working, and use words and labels from our lessons (centre of rotation, angle of rotation, clockwise, anticlockwise, order of rotational symmetry). Draw neat sketches where a prompt asks for them. Use a ruler, compass and protractor where needed.

#### Part 1: What changes and what stays the same?

The minute hand of the University of Nairobi clock was observed at 12:00, 12:15, 12:30 and 12:45. The hand pointed in a different direction each time, yet it looked the same. (a) Name the type of motion the hand is performing and say how it differs from a translation and from a reflection. (b) Say which point is the centre of rotation, and explain why it is the only point that does not move. (c) List what changes and what stays the same about the hand during the motion, and give a reason for each. (d) Explain how a compass and a protractor in the four-step construction method represent features of the clock.

Write your answer here:

### Part 2: Predicting positions with coordinate rules

Model the minute hand as a line segment on a coordinate grid with the clock centre at the origin  $O(0, 0)$ . The points  $A(0, 5)$ ,  $B(0, 3)$  and  $C(0, 1)$  lie on the hand at 12:00. The hand turns  $90^\circ$  clockwise every 15 minutes.

| Time  | Turn from 12:00       | A      | B      | C      |
|-------|-----------------------|--------|--------|--------|
| 12:00 | $0^\circ$             | (0, 5) | (0, 3) | (0, 1) |
| 12:15 | $90^\circ$ clockwise  | ?      | ?      | ?      |
| 12:30 | $180^\circ$           | ?      | ?      | ?      |
| 12:45 | $270^\circ$ clockwise | ?      | ?      | ?      |

(a) Write down the coordinate rule for each turn:  $90^\circ$  clockwise,  $180^\circ$  and  $270^\circ$  clockwise. (b) Use the rules to complete the table. (c) Using point A at 12:15, show by calculation that its distance from the centre has not changed. (d) State the position of A at 1:00 and explain why.

Write your answer here:

### Part 3: Finding the centre and angle of rotation

Suppose you did not know where the clock's pivot was. You are only told that the tip of the hand moved from  $A(0, 5)$  at 12:00 to  $A'(5, 0)$  at 12:15, and then to  $A''(0, -5)$  at 12:30. (a) Describe, step by step, how you would use a ruler and compass to find the centre of rotation from these positions. (b) Plot  $A$ ,  $A'$  and  $A''$  on a coordinate grid and carry out your construction on the grid. Use it to state the coordinates of the centre. (c) Explain why the centre must lie on the perpendicular bisector of each object–image segment. (d) How would you measure the angle of rotation and check your answer?

*Write your answer here:*

#### Part 4: Rotational symmetry

- (a) What does it mean for a plane figure to have rotational symmetry of order  $n$ , and how do you work out the smallest angle of rotation that maps it onto itself? (b) State the order of rotational symmetry of a regular hexagon (like the weave pattern of a Maasai kiondo basket), a rectangle, and the letter S. (c) The clock face at the University of Nairobi has 12 equally spaced marks. State its order of rotational symmetry and the angle between each coincident position. (d) A triangular prism is rotated about its longitudinal axis. State the order of rotational symmetry it has about that axis and explain why.

*Write your answer here:*

### Part 5: Answering the driving question

**Write a complete, evidence-based answer to the driving question: When the minute hand of the University of Nairobi clock completes its journey around the clock face, what mathematical rules govern exactly where every point on the hand ends up, and how can we use those rules to rotate any shape to any position, on paper and on a grid? Your answer should (i) state the properties of rotation, (ii) explain how to find the image on plain paper and on the Cartesian plane, (iii) explain how to find the centre and angle when they are unknown, and (iv) give one example of rotation from your school, home or a Kenyan place, stating its centre and an angle or order of symmetry. Finally, say how your thinking has changed since your first sketch in Lesson 1.**

Write your answer here:

| RUBRIC                                            |                                                                                                                                                            |                                                                                             |                                                                                                     |
|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Criterion                                         | Excellent (4)                                                                                                                                              | Proficient (3)                                                                              | Developing (1–2)                                                                                    |
| Properties of rotation and link to the phenomenon | Correctly names the centre, angle, direction and preserved distances and shape, and clearly links each to the Nairobi clock and to the construction steps. | States most properties correctly with some link to the clock; minor gaps in explaining why. | Names rotation but confuses it with translation or reflection, or lists properties without reasons. |
| Applying coordinate rules                         | Uses all rules correctly to complete the table, verifies distance preservation with calculation, and explains the $360^\circ$ return.                      | Completes most of the table correctly; minor sign errors or incomplete distance check.      | Uses wrong rules or direction, or cannot calculate image coordinates.                               |
| Finding centre and angle of rotation              | Gives an accurate perpendicular bisector construction, finds the correct centre, justifies why the centre is equidistant, and verifies the angle.          | Describes the method and finds the centre with minor errors in reasoning or verification.   | Cannot locate the centre or does not use perpendicular bisectors.                                   |
| Rotational symmetry                               | Gives correct orders and angles for all figures and the solid, with the $360^\circ \div n$ reasoning.                                                      | Most orders correct; reasoning is partial.                                                  | Orders incorrect or no use of $360^\circ \div n$ .                                                  |
| Synthesis and real-life application               | Writes a coherent, evidence-based answer covering all four points, gives a valid Kenyan example, and reflects on                                           | Answer covers most points with a reasonable example; reflection is brief.                   | Answer is incomplete or lacks evidence, example or reflection.                                      |

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|  | how their thinking changed. |  |  |
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